MOYI TIAN

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EDUCATION

Brown University, Providence, RI — *September 2019 - May 2024*
Ph.D. in Applied Mathematics
Thesis Advisor: Dr. Björn Sandstede, Division of Applied Mathematics
Thesis: Patterns in Network Dynamics
MS, Applied Mathematics — *May 2021*

Dickinson College, Carlisle, PA — *September 2015 - May 2019*
B.S. in Mathematics & Physics
Phi Beta Kappa Honors, Honors in Mathematics, *Summa Cum Laude*

POSITIONS

Academic

University of Colorado Boulder, Boulder, CO — *August 2024 - Present*
Postdoctoral Associate
Mentors: Nancy Rodríguez (primary), David Bortz (secondary)

Industry

Los Alamos National Laboratory, Los Alamos, NM — *September 2023 - November 2023*
Applied Mathematics Graduate Student Intern

SAS Institute, Inc, Remote — *May 2023 - August 2023 (full-time)*
Machine Learning Graduate Intern — *& January 2024 - May 2024 (part-time)*

Oak Ridge National Laboratory, Oak Ridge, TN — *June 2022 - August 2022*
NSF Mathematical Sciences Graduate Intern

RESEARCH INTERESTS

Nonlinear dynamical systems, network science, pattern formation, and data-driven modeling and scientific machine learning, with a focus on collective and social dynamics, including human–AI interactions.

PUBLICATIONS

†: undergraduate student mentored by me.

Published Articles

2026 **M Tian**, PJ Brantingham, and N Rodríguez, *Modelling the Spillover from Online Engagement to Offline Protest: Stochastic Dynamics and Mean-Field Approximations on Networks*, Journal of Complex Networks **14** (2026), no. 2, cnaf057

2023 **M Tian** and P Moriano, *Robustness of Community Structure under Edge Addition*, Physical Review E **108** (2023), 054302

2021 **M Tian**, JJ Bramburger, and B Sandstede, *Snaking Bifurcations of Localized Patterns on Ring Lattices*, IMA Journal of Applied Mathematics **86** (2021), no. 5, 1112–1140

2019 H Mhiri, **M Tian**, E Wynne, S Jones, A Mareno, and LQ English, *An Experimental Survey of Chaos and Symmetry Breaking in Coupled and Driven Logistic Maps*, European Journal of Physics **40** (2019), no. 6, 065802

Preprints

2026 W Thompson, **M Tian**, L Huynh, A Wiedemann, V Modisette, PS Chodrow, and HZ Brooks, *Learning Social Influence with Unobserved Attention*, arXiv:2610.12616 (2026)

2026 B Huynh[†], **M Tian**, and N Rodríguez, *Group Dynamics of Engagement with AI Topics on Bluesky*, arXiv:2609.36011 (2026) (project conceived and designed by me)

2026 **M Tian**, DA Messenger, V Dukic, N Rodríguez, and DM Bortz, *Learning Effective Models from Network Dynamics Data with Multiple Initial Conditions Using Weak Form SINDy*, arXiv:2605.30432 (2026)

2026 **M Tian**, G Mohler, PJ Brantingham, and N Rodríguez, *Learning Dynamics from Online-Offline Systems of LLM Agents*, arXiv:2602.23437 (2026)

In Preparation

- **M Tian** and B Sandstede, *Bifurcation from Homogeneous Branches on Ring Lattices.*

Software

2026 **M Tian**, *WENDy-Constrained: Weak-Form Parameter Estimation with a Structure Matrix and an Offset Vector for Constrained ODE Coefficients*, Zenodo (2026), v1.0.0. DOI: 10.5281/zenodo.22006935

SELECTED RECOGNITION & MEMBERSHIPS

Awards

2026 **SIAM Conference Travel Support**, Society for Industrial and Applied Mathematics
Travel support to attend SIAM Conference on the Life Sciences

2026 **Dynamics Days 2026 Travel Awards**, Dynamics Days US 2026
Travel support to attend Dynamics Days US 2026 conference

2025 **Dynamics Days 2025 Travel Awards**, Dynamics Days US 2025
Travel support to attend Dynamics Days US 2025 conference

2024 **SIAM Student Chapter Certificate of Recognition**, Society for Industrial and Applied Mathematics
In recognition of outstanding efforts and accomplishments on behalf of the SIAM Chapter at Brown University

2023 **SIAM Student Travel Awards**, Society for Industrial and Applied Mathematics
Support students to attend and present at SIAM conferences

2019 **James Fowler Rusling Prize**, Dickinson College
Presented to a member of the senior class whose scholarly achievements have been judged most superior by the All-College Committee on Academic Program and Standards

Honors

Phi Beta Kappa Honor Society
Pi Mu Epsilon National Honorary Mathematics Society
Sigma Pi Sigma National Physics Honor Society
Alpha Lambda Delta Honor Society

Memberships

2025 - Present American Physical Society
2025 - Present Association for Women in Mathematics
2019 - Present Society for Industrial and Applied Mathematics
2019 - 2024 American Mathematical Society

SELECTED PRESENTATIONS

Talks

2026 **27th International Symposium on Mathematical Theory of Networks and Systems (MTNS 2026)**
Learning Models from Online–Offline Social Dynamics on Networks Using WSINDy, Waterloo, ON, Canada, August, 2026

2026 **2026 SIAM Annual Meeting (AN26)**
Connecting Models with Data for Coupled Online–Offline Social Systems, Cleveland, OH, July, 2026

2026 **SIAM Conference on the Life Sciences (LS26)**
Collective Behavior in Social Mobilization: Multilayer Online–Offline Dynamics on Networks, Cleveland, OH, July, 2026

2026 **APS Global Physics Summit 2026**
Learning Models from Network Dynamics Data using Weak Form SINDy, Denver, CO, March, 2026

2026 **Dynamics Days US 2026**
Modeling and Learning Dynamics of Online–Offline Social Systems on Networks, Tucson, AZ, January, 2026

2025 **ICERM Workshop on Patterns, Dynamics, and Data in Complex Systems**
Localized Patterns and Modeling Dynamics on Networks, Providence, RI, January, 2025

2024 **2024 Joint Mathematics Meetings (JMM 2024)**
Inferring Interaction Kernels for Stochastic Agent-Based Opinion Dynamics, San Francisco, CA, January, 2024

2023 **SIAM Conference on Applications of Dynamical Systems (DS23)**
Localized Patterns on Graphs, Portland, OR, May, 2023

2022 **SIAM Workshop on Network Science (NS22)** (virtual) - Lightning Talk
How Robust are Communities in Temporal Networks? A Comparative Analysis Using Community Detection Algorithms, September, 2022

2021 **Brown / BU / UMass Dynamics & PDE Seminar**
Snaking Bifurcations of Localized Patterns, University of Massachusetts Amherst, Amherst, MA, November, 2021

Posters

2026 **NetSci 2026**
Modeling and Learning Dynamics of Online–Offline Social Systems on Networks, Boston, MA, June, 2026

2025 **SIAM Conference on Applications of Dynamical Systems (DS25)**
Modeling Coupled Online and Offline Dynamics of Protesting Activity on Networks, Denver, CO, May, 2025

2025 **Dynamics Days US 2025** - Flash Talk & Poster
Localized Patterns on Ring Lattices, Denver, CO, January, 2025

2024 **AWM Workshop at SIAM Annual Meeting 2024**
Efficiently Learning Models of Networks, Spokane, WA, July, 2024

2024 **2024 Dynamics Days US**
Efficient Learning of Models for Temporal Networks, Davis, CA, January, 2024

2023 **Mathematical Opportunities in Digital Twins (MATH-DT) Workshop**
Community Robustness under Edge Addition in Synthetic and Empirical Temporal Networks, George Mason University, Fairfax, VA, January, 2023

2023 **Dynamics Days US 2023** (virtual)
Community Robustness in Temporal Networks under Edge Addition, January, 2023

2021 **SIAM Conference on Applications of Dynamical Systems (DS21)** (virtual)
Snaking Bifurcations of Localized Patterns on Ring Lattices, May, 2021

MENTORING & SERVICE

Mentoring

Summer 2026 **Undergraduate Research Mentoring**, University of Colorado Boulder
Independently mentored two undergraduate students on separate projects involving (1) large-scale simulation and analysis of reaction–diffusion dynamics on random networks and (2) collection, processing, and analysis of Bluesky activity data.

Fall 2021 **Brown Division of Applied Mathematics - Directed Reading Program**, Brown University
Mentored an undergraduate student on mathematical optimal control theory and applications

Summer 2020 **5-week Undergraduate Program in Experimental Math**, Brown University
Supervised a team of 3 undergraduate students on studying the dynamics and patterns of graphing fleas

2019 - 2020 **Brown Applied Math Undergraduate/Graduate Mentoring Program**
Mentored an undergraduate student with regards to study plans and academic/career goals

Service

AY 2026–27 **Co-Organizer, APPM Postdoctoral Seminar**, University of Colorado Boulder Department of Applied Mathematics; coordinate speaker invitations, scheduling, and seminar logistics

July 2026 **Co-Organizer, 3-Part Minisymposium**, 2026 SIAM Annual Meeting (AN26)
Advances in Modeling and Inference for Dynamics on Networks, Cleveland, OH

2021 - 2023 **APMA Diversity, Equity, and Inclusion Committee**, Brown University
Attended bi-weekly meetings and developed plans and projects to improve climate and increase diversity and inclusion in the Division of Applied Mathematics

2021 - 2023 **Brown SIAM Student Chapter Executive Board**, Brown University
Kept record for organizational plans and helped facilitate on-campus events

2021 - 2023 **Sheridan Center Departmental Graduate Student Liaison**, Brown University
Maintained communications and forwarded events information between the Sheridan Center for Teaching and Learning and the Division of Applied Mathematics

TEACHING

Experience

Fall 2025 **Introduction to Differential Equations with Linear Algebra**, Instructor, University of Colorado Boulder

Spring 2021	**Applied Ordinary Differential Equations**, Teaching Assistant, Brown University
Fall 2020	**Operations Research: Deterministic Models**, Teaching Assistant, Brown University
Fall 2018	**Single Variable Calculus**, Teaching Assistant, Dickinson College
Fall 2017	**Introduction to Calculus**, Teaching Assistant, Dickinson College
Fall 2016	**Single Variable Calculus**, Teaching Assistant, Dickinson College

Pedagogical Training

Fall 2020 **Sheridan Center Certificate I: Reflective Teaching**, Brown University
Introductory seminar in a cross-disciplinary setting that helps develop and refine fundamental, evidence-based teaching skills and strategies

Tutoring

2017 - 2019 **Quantitative Reasoning Center**, Dickinson College
Tutored college entry-level math, physics and economics students/classes

2016 - 2019 **Math Help Room**, Dickinson College
Tutored walk-in students from college entry-level mathematics classes

SELECTED WORKSHOPS & PROGRAMS

2025	IMSI Workshop: UQ and Trustworthy AI Algorithms for Complex Systems and Social Good, Chicago, IL, March 3 - 7, 2025
2024	IPAM Workshop IV: Modeling Multi-Scale Collective Intelligences, Los Angeles, CA, November 18 - 22, 2024
2023	American Mathematical Society Mathematics Research Communities 2023 on Complex Social Systems, Java Center, NY, June 18 - June 24, 2023
2023	Women in Network Science Collabathon, group project on topic modeling, Northeastern University, May 1 - May 5, 2023
2022	Fall 2022 Data Science Boot Camp, group project on English language proficiency evaluation model, the Erdős Institute, September - December, 2022 (virtual)
2022	OLCF Summer Hands-On High Performance Computing Course, Certificate of Completion, Oak Ridge National Laboratory, July 2022 (virtual)
2022	May 2022 Data Science Boot Camp, group project on predicting chocolate ratings with feature engineering, the Erdős Institute, May 9 - June 4, 2022 (virtual)
2020	IMSD Module: Introduction to Statistical Analysis of Data 2020, Brown University, November 5, 12 and 19, 2020 (virtual)
2020	IMSD Module: Scientific Presentations, Brown University, June 4, 5 and 11, 2020 (virtual)
2018	Budapest Semesters in Mathematics Program, completed with honors, Budapest, Hungary, Summer 2018